THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 12 & 13: **How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Non-living components of an ecosystem are: 1

(a) producers **(b)** consumers
(c) biotic components **(d)** abiotic components

2. A pond is an example of an: 1

(a) grassland **(b)** desert ecosystem
(c) aquatic ecosystem **(d)** terrestrial ecosystem

3. Organisms that eat only animals are: 1

(a) herbivores **(b)** decomposers **(c)** producers **(d)** carnivores

4. Decomposers help the ecosystem by: 1

(a) hunting prey **(b)** producing sunlight
(c) making oxygen **(d)** recycling nutrients

5. In the chain grass → grasshopper → frog → snake, the secondary consumer is: 1

(a) grass **(b)** frog **(c)** grasshopper **(d)** snake

6. The habitable zone is also called the: 1

(a) Hill zone **(b)** Roche zone **(c)** Goldilocks zone **(d)** Van Allen zone

7. The ozone layer protects life from: 1

(a) infrared only **(b)** visible light
(c) sound **(d)** harmful ultraviolet rays

8. Venus is hotter than Mercury due to: 1

(a) a strong greenhouse effect **(b)** no atmosphere
(c) ozone layer **(d)** distance from Sun

9. The biosphere consists of: 1

(a) life **(b)** air **(c)** rocks **(d)** water

10. Sexual reproduction involves: 1

(a) only cutting stems **(b)** two parents and fusion of gametes
(c) a single parent **(d)** no gametes

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define commensalism with an example. 2

12. Define parasitism with an example. 2

13. Why is reproduction essential? 2

14. Differentiate asexual and sexual reproduction in one point each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How do human activities threaten ecosystems? Give three examples. 3

16. Why does sexual reproduction help living things survive better? 3

17. Explain the role of national parks and sanctuaries. 3

OR

Name any three protected areas of India and their states. 3

18. List three ways human activities harm the Earth. 3

OR

Why do small changes in temperature matter for life on Earth? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain how human activities affect ecosystems and what individuals can do. 5

OR

Describe the work of conservation: national parks, sanctuaries, biosphere reserves and community areas. 5

20. Describe how soil, rocks, minerals, plants, animals and microbes sustain life together. 5

OR

Explain with an example why balance in nature is important. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND